

The Frank–Sabin weak Schatten endpoint: dimension one is false, dimensions $d \geq 2$ remain open

Literature-identification note for arXiv:1609.08388

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Status. Literature-implied answer (dimension-one counterexample). The endpoint is false for $d = 1$. No answer is claimed for $d \geq 2$, where a 2025 source still records the conjecture as open.

1 The source question

Let

$$S = \{(\omega, \xi) \in \mathbb{R} \times \mathbb{R}^d : \omega = -|\xi|^2\}, \quad T_S = \mathcal{E}_S \mathcal{E}_S^*.$$

Frank and Sabin prove mixed-norm Schatten estimates for $q > d + 1$ and, in Remark 5 on PDF page 6 of [1], isolate the remaining endpoint. In their notation the proposed estimate is

$$\|\overline{W} T_S W\|_{\mathfrak{S}^{d+1, \infty}(L^2(\mathbb{R}^{d+1}))} \lesssim \|W\|_{L_t^{2(d+1)} L_x^{d+1}}^2. \quad (1)$$

They state that the corresponding strong $\mathfrak{S}^{d+1}$ estimate is false and that (1) was completely open. The source evidence is reproduced in `figures/source_endpoint_page6.png`.

2 The exact dual formulation

Put $V = |W|^2$ and define the operator on $L^2(\mathbb{R}^d)$

$$A_V = \int_{\mathbb{R}} e^{it\Delta} V(t, \cdot) e^{-it\Delta} dt.$$

The usual AA^*/A^*A correspondence identifies the nonzero spectral data in (1) with that of A_V . For $r = d + 1$ and $r' = (d + 1)/d$, Lorentz–Schatten duality gives $(\mathfrak{S}^{r', 1})^* = \mathfrak{S}^{r, \infty}$. Moreover,

$$\mathrm{Tr}(\Gamma A_V) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} V(t, x) \rho_{e^{-it\Delta} \Gamma e^{it\Delta}}(x) dx dt. \quad (2)$$

Mixed Lebesgue duality therefore makes (1) equivalent to

$$\left\| \sum_j \lambda_j |e^{it\Delta} f_j|^2 \right\|_{L_t^{(d+1)/d} L_x^{(d+1)/(d-1)}} \lesssim \|\lambda\|_{\ell^{(d+1)/d, 1}} \quad (3)$$

for orthonormal (f_j) . To pass from the diagonal source formulation to an arbitrary complex dual test V , decompose its real and imaginary parts into positive and negative parts; each nonnegative

part is $|W|^2$. Since $d + 1 > 1$, the weak Schatten ideal is normable, so this decomposition loses only an absolute constant.

For $d = 1$, (1) and (3) become, respectively,

$$\|\overline{W}T_S W\|_{\mathfrak{S}^{2,\infty}} \lesssim \|W\|_{L_t^4 L_x^2}^2, \quad \left\| \sum_j \lambda_j |e^{it\Delta} f_j|^2 \right\|_{L_t^2 L_x^\infty} \lesssim \|\lambda\|_{\ell^{2,1}}.$$

3 The later counterexample

Bez, Hong, Lee, Nakamura, and Sawano formulate (3) as their Conjecture 1.3. Their Theorem 5.3 on PDF page 27 states that, when $d = 1$, the induced endpoint velocity-averaging estimate

$$\left\| \int_{\mathbb{R}} f(x - tv, v) dv \right\|_{L_t^2 L_x^\infty} \lesssim \|f\|_{L^{2,1}(\mathbb{R}^2)} \quad (4)$$

fails, and hence their conjecture is false [2]. Proposition 5.1 of the same paper proves by semiclassical Weyl quantization that (3) would imply (4). The proof of Theorem 5.3, continued on page 28, tests (4) on shrinking neighborhoods of a planar measure-zero Kakeya set. Thus (3) is false for $d = 1$, and the duality in (2) proves:

Conclusion. There is no constant C such that

$$\|\overline{W}T_S W\|_{\mathfrak{S}^{2,\infty}} \leq C \|W\|_{L_t^4 L_x^2}^2$$

for every W . This is a full negative answer to the source endpoint in dimension one.

4 Scope and current status

The Kakeya proof uses the endpoint spatial norm L_x^∞ , special to $d = 1$. It does not settle higher dimensions. Feng, Song, and Wu restate (3) as Conjecture 1.6 and, immediately after it on PDF page 6, write that it is false for $d = 1$ but remains open for $d \geq 2$ [3]. Their status page is reproduced in `figures/current_status_page6.png`.

The literature search checked the exact weak-Schatten and restricted orthonormal endpoint phrases, the earlier strict-range theorem arXiv:1404.2817, the 2017 counterexample, and the 2025 critical-summability paper. No later all-dimensional proof or counterexample was found. Eight focused upgrade routes are recorded separately; none overcame the surviving mixed-norm/Kakeya endpoint for $d \geq 2$.

References

- [1] R. L. Frank and J. Sabin, *The Stein–Tomas inequality in trace ideals*, Séminaire Laurent Schwartz—EDP et applications (2015–2016), Exp. XV; arXiv:1609.08388.
- [2] N. Bez, Y. Hong, S. Lee, S. Nakamura, and Y. Sawano, *On the Strichartz estimates for orthonormal systems of initial data with regularity*, arXiv:1708.05588, Theorem 5.3.
- [3] G. Feng, M. Song, and H. Wu, *Strichartz estimates involving orthonormal systems at the critical summability exponent*, arXiv:2507.14974, Conjecture 1.6.